

E-Commerce Conversion Prediction

Methodology Report • Summer Analytics 2026 • IIT Guwahati C&A Club

Objective

We aimed to predict if a visitor would buy or browse. I treated this as a binary classification task and used the F1 Score to measure success, ensuring we caught actual buyers without too many false alarms.

Dataset Overview

- Train set: 10,000 rows, 14 columns — used to train the model
- Public test: 3,000 rows with labels — used for validation and threshold tuning
- Private test: 3,000 rows without labels — final submission target
- Class split: ~69% not converted, ~31% converted — moderately imbalanced

We cleaned the data by predicting missing values through iterative imputation and capping session outliers

Exploratory Data Analysis

Initial analysis revealed significant missing data in Age, Income, and Time_On_Site. We also identified a major outlier in session duration. Key drivers for conversion included page and product views, session length, and campaign type.

Feature Engineering

Eight new features were created across two rounds of engineering to give the model stronger behavioural signals:

- Products_per_Page — browsing focus: how many products viewed per page
- Time_per_Page — engagement depth: time spent per page visited
- Time_per_Product — consideration score: time spent per product viewed
- Discount_x_PrevPurchase — interaction between discount visibility and purchase history
- Income_per_Age — a simple wealth approximation
- High_Engagement — binary flag: above-median products viewed AND time on site
- Age_Group — age binned into four life stages: 0-25, 25-35, 35-50, 50+
- Activity_Score — overall behavioural intensity: pages + products + previous purchases

Modelling Approach

We evolved our model from a simple Random Forest (F1: 0.31) to tuned XGBoost and LightGBM models. The best performance came from a soft-voting ensemble of these gradient-boosted trees, which significantly improved our ability to identify actual buyers.

Threshold Tuning

Instead of the default 0.5 decision threshold, thresholds between 0.10 and 0.90 were evaluated on the public test set at 0.02 intervals. The threshold that maximised F1 Score was selected — this meaningfully boosted recall for the converted class without sacrificing too much precision.

Final Model

The final ensemble (XGBoost + LightGBM) was retrained on all available labelled data — train.csv and public_test.csv combined (13,000 rows, 20 features) — to maximise available signal before generating predictions for private_test.csv. Both models used: 500 estimators, learning rate 0.03, max depth 5, subsample 0.8, colsample_bytree 0.8, and scale_pos_weight to handle class imbalance.

Results

Model Version	Evaluated On	F1 Score
Baseline — Random Forest	Public Test	0.3084
XGBoost + Feature Engineering + Tuning	Public Test	0.5381
XGBoost + Iterative Imputer + New Features	Public Test	0.5407
LightGBM alone	Public Test	0.5442
XGBoost + LightGBM Ensemble	Public Test	0.5454

Private test F1 score will be revealed after submission results are released by the organizers. Best validated score on public test: 0.5454 using the soft voting ensemble.

Tools & Libraries

Python 3 • pandas • NumPy • scikit-learn • XGBoost • LightGBM • Matplotlib • Jupyter Notebook (Anaconda)